

# JOSHUA OGUNDAIRO

Department of Statistics, Federal University of Technology, Akure, Nigeria

ogundairosta154470@futa.edu.ng · +234 807 367 5336 · [oluwadamilarejoshua.github.io](https://oluwadamilarejoshua.github.io) · [GitHub](#)

## RESEARCH INTERESTS

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Bayesian Computation; Time Series and State-Space Models; Likelihood Approximation and Sequential Monte Carlo; Volatility Modelling; Financial Econometrics.

## EDUCATION

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- 2024 – 2026     **MTech in Statistics**, Federal University of Technology, Akure (FUTA). CGPA 4.53/5.0; Distinction (expected October 2026)  
Thesis: *Likelihood Approximation Choices and the Performance of Markov-Switching GARCH Models*. Advisor: Prof. Oluwadare O. Ojo
- 2015 – 2021     **BTech in Statistics**, FUTA. CGPA 3.76/5.0; Second Class Upper Division  
Selected coursework: Real Analysis I (B), Real Analysis II (A), Probability Distributions I (A), Stochastic Processes I (A), Statistical Inference I (A), Econometrics (A), Time Series Analysis (A), Bayesian Inference (A), Multivariate Analysis (B)

## RESEARCH COMPUTING

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- Methods        Bayesian computation (Stan/RStan, HMC/NUTS,  $\hat{R}$ , LOO-CV); likelihood approximation (collapsing, quadrature, particle filtering); Monte Carlo experiment design; GARCH-family, Markov-switching, and VAR/VECM models
- Software       R, Python, C++ (via Cython), SQL, EViews, Stata,  $\text{\LaTeX}$ . Open-source: [PSTR-VECM](#) panel cointegration tests in R

## RESEARCH PAPERS

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### *Working Papers*

“Likelihood Approximation Choices and the Performance of Markov-Switching GARCH Models: Evidence from Thirty Global Financial Assets,” **J. Ogundairo** and O. O. Ojo. MTech thesis. *Accepted for poster presentation, CFE-CMStatistics 2026, Berlin*.

Develops a kurtosis-based framework for choosing among four likelihood approximations for Markov-switching GARCH models (Gray’s collapsing, plug-in, Gauss-Hermite quadrature, particle filter). Under standard optimisation the particle filter trails the analytical approximations by 2,000–4,000 log-likelihood units, while plug-in dominates in every kurtosis category.

*My role:* Designed the study and its 220-case experiment, built a C++/Cython likelihood engine (36-fold speed-up), ran all estimation, and wrote the thesis.

“Impacts of GDP and Debt Servicing on Exchange Rate Volatility of the Nigerian Naira: A Bayesian EGARCH-X Approach,” **J. Ogundairo**, A. A. Adewale, and O. O. Ojo. *In review for bid at the Oxford CSAE Conference, March 2027*.

Places macro covariates in the variance equation of a Bayesian EGARCH(1,1)-X, after structural-break tests rule out mean effects, and estimates it by NUTS (4 chains,  $\hat{R} \leq 1.01$ , LOO-CV): debt servicing raises conditional variance by about 32% per s.d. (95% CrI [0.163, 0.414]) and GDP growth lowers it by about 13%.

*My role:* Conceived the study, assembled the data, built and estimated the Bayesian model in RStan, and wrote the paper as lead author.

## Publications

“Forecasting of Crude Oil Prices: A Comparative Time Series Prediction Technique,” O. O. Ojo, A. A. Adewale, and **J. Ogundairo**. *Studies of Applied Economics*, 44(3), 2026.

Compares GARCH, Support Vector Regression, and LSTM forecasts of daily WTI and Brent prices, showing that the best model depends on forecast horizon and market regime.

*My role:* Implemented the GARCH, SVR, and LSTM models in R and contributed to the comparative analysis.

“Energy-Growth Dynamics in the Presence of Income of 5 Oil-Exporting African Countries: A Time Series Approach,” **J. Ogundairo**, P. T. Oluwadare, and L. Di Gennaro Splendore. *Journal of the Royal Statistical Society – Nigeria Group*, 1(2), 19–43, 2024.

Uses unit-root, Engle-Granger cointegration, VAR/VECM, and Granger causality analysis to show that the energy-growth relationship takes four different causal forms across five neighbouring oil exporters (1980–2021).

*My role:* Conceived the study, assembled the WDI and US EIA data, carried out all time series analysis, and wrote the paper as lead author.

## OTHER RESEARCH PRESENTATIONS

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| 2025 | “Levels and Determinants of Anxiety and Depression among Undergraduate Finalists in FUTA” (with A. Aladeniyi, K. Baughman, O. Arigbede, and T. Amusa). <i>APHA 2025 Annual Meeting</i> , Washington, D.C. |
| 2025 | “Macroeconomic Drivers of Exchange Rate Volatility in Nigeria: Evidence from a GARCH-M Approach.” <i>OAUSTECH International Conference</i> , Okitipupa.                                                   |

## HONOURS AND AWARDS

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| 2026 | 1st Place, EGRA (EHA Group-wide Research Accelerator), inaugural round, eHealth Africa, for a spatial-epidemiology study. |
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## TEACHING EXPERIENCE

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| 2024 – 2025    | <b>Research and Teaching Assistant</b> , Department of Statistics, FUTA <ul style="list-style-type: none"><li>– Lectured STA224 (Statistics for Environmental and Biological Sciences).</li><li>– Graded continuous assessments: assignments, class work, and participation.</li><li>– Trained student interns and graduates in R, Python, SQL, Stata, and SPSS.</li></ul> |
| 2020 – 2021    | <b>Senior Statistical Analyst and Lead Trainer</b> (voluntary), FUTA-LISA. Leads statistical                                                                                                                                                                                                                                                                               |
| 2024 – present | analysis for university researchers; has trained 150+ learners in R and data analytics.                                                                                                                                                                                                                                                                                    |

## RESEARCH AND PROFESSIONAL EXPERIENCE

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| 2025 – present | <b>Senior Coordinator, Data Scientist</b> , eHealth Africa, Abuja <ul style="list-style-type: none"><li>– Leads a portfolio of five spatial-epidemiology studies (two under review).</li><li>– Leads a pre/post impact evaluation of a carbon-reduction initiative.</li><li>– Builds data pipelines on Databricks; mentors interns in research.</li></ul> |
| 2025           | <b>Principal Data Analyst</b> (contract), Greenplinth Africa, Lagos <ul style="list-style-type: none"><li>– Designed data collection for a large clean-cookstove intervention.</li><li>– Trained enumerators on survey instruments and implementation.</li><li>– Analysed field data to support carbon-emission reduction decisions.</li></ul>            |